

MANNVILLE AGCM, AB, Canada

WMO: 710470

Lat: 53.4556N

Lon: 111.2553W

Elev: 656

StdP: 93.69

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.1	-28.3	-34.6	0.2	-31.0	-31.5	0.2	-28.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	27.8	16.9	26.1	16.4	24.5	15.9	19.4	25.1	18.1	23.7	17.0	22.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.3	22.6	15.9	12.2	21.1	14.8	11.4	19.8	58.2	24.9	53.9	23.6	50.3	22.7	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	-34.5	30.8	1.5	1.2	-35.6	31.7	-36.5	32.4	-37.3	33.1	-38.4	34.0		
			-34.8	21.6	1.6	1.3	-36.0	22.5	-36.9	23.2	-37.8	23.9	-39.0	24.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-11.3	-12.1	-5.4	3.0	11.2	15.0	16.8	16.1	11.5	3.9	-5.1	-12.6
	DBStd	12.56	8.74	8.01	7.54	5.60	4.66	3.16	2.58	3.10	4.81	5.06	7.09	8.20
	HDD10.0	3414	660	619	478	218	42	4	0	1	38	198	453	701
	HDD18.3	5763	918	853	736	461	223	107	62	82	210	448	703	959
	CDD10.0	731	0	0	0	7	79	153	211	189	83	8	0	0
	CDD18.3	39	0	0	0	0	2	6	15	12	4	0	0	0
	CDH23.3	604	0	0	0	3	73	82	169	173	102	2	0	0
	CDH26.7	103	0	0	0	0	11	8	28	28	29	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	414	19	12	16	30	44	70	76	58	34	18	19	17
	PrecMax	562	61	23	37	104	93	139	166	118	78	46	76	32
	PrecMin	209	4	3	5	3	5	14	25	12	1	2	2	4
	PrecStd	80	12	6	8	20	23	29	30	23	21	10	16	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	5.4	11.8	22.4	28.3	27.9	29.5	29.4	30.1	21.7	12.9	5.5
		MCWB	2.7	2.2	6.0	10.1	13.5	16.4	19.5	18.2	15.9	12.1	6.6	1.6
	2%	DB	4.5	3.2	8.0	18.6	25.5	25.7	27.0	27.0	26.3	16.7	8.3	3.5
		MCWB	1.6	0.8	3.6	8.5	12.3	15.5	19.0	17.6	14.7	9.4	4.3	0.6
	5%	DB	3.0	1.5	5.3	15.2	22.8	23.7	25.2	25.3	23.2	14.0	5.3	1.0
		MCWB	0.7	-0.4	2.3	6.8	11.5	14.9	18.1	17.3	13.7	7.7	2.4	-1.0
	10%	DB	1.1	-0.3	3.4	11.9	20.4	21.9	23.4	23.3	20.0	11.4	3.2	-1.5
		MCWB	-0.8	-1.6	1.0	5.2	10.4	13.9	16.9	16.3	12.4	6.7	1.1	-3.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.6	6.6	10.8	14.6	18.8	21.6	20.6	17.1	12.9	7.1	2.2
		MCDB	5.7	5.2	11.1	22.0	25.2	25.3	27.0	26.1	26.7	20.8	11.6	5.0
	2%	WB	1.9	1.0	3.8	8.6	13.2	17.1	20.0	19.0	15.5	10.6	4.5	0.6
		MCDB	4.2	2.7	7.4	17.3	22.8	23.0	25.8	24.8	24.7	15.0	7.9	3.2
	5%	WB	0.8	-0.3	2.3	7.1	12.2	15.9	18.6	17.8	14.2	8.7	2.8	-1.0
		MCDB	2.9	1.4	5.0	14.4	20.8	21.7	23.8	23.6	22.2	13.0	5.0	1.0
	10%	WB	-0.8	-1.7	1.2	5.5	11.2	14.9	17.5	16.8	12.8	7.0	1.3	-3.1
		MCDB	1.1	-0.2	3.3	11.1	19.2	20.3	22.4	22.3	19.1	11.3	3.2	-1.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.0	8.6	10.5	12.5	11.0	11.1	11.9	12.3	9.7	7.6	7.6
		MCDBR	9.4	10.0	9.9	16.1	16.0	13.8	14.0	15.3	17.6	14.1	10.1	10.1
		MCWBR	7.3	8.3	6.5	8.3	6.7	6.1	7.5	7.4	7.7	7.9	6.8	8.0
	5% WB	MCDBR	9.0	9.5	9.6	15.3	14.4	12.0	12.9	13.7	16.5	13.0	9.5	10.3
		MCWBR	7.2	8.1	6.5	8.2	6.4	6.1	7.4	7.2	7.6	7.6	6.7	8.5
(40) Clear-Sky Solar Irradiance	taub		0.248	0.284	0.295	0.331	0.350	0.368	0.364	0.376	0.312	0.276	0.249	0.219
	taud		2.316	2.244	2.347	2.361	2.356	2.334	2.370	2.312	2.475	2.548	2.446	2.359
	Ebn at Noon		750	818	896	901	897	881	878	842	870	842	757	735
	Edh at Noon		56	84	95	108	114	118	112	112	84	62	49	43
(44) All-Sky Solar Radiation	RadAvg		0.98	2.04	3.55	4.69	5.83	5.95	6.15	5.02	3.53	2.04	0.98	0.73
	RadStd		0.10	0.15	0.28	0.41	0.41	0.43	0.34	0.37	0.39	0.29	0.11	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page